Hope Foundation's International Institute of Information Technology, Pune	
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Course Name: LP-II	**Course Code: 314458**
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List of Laboratory Assignments	

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2	Write a JavaScript Program to get the user registration data and push to array/local storage with AJAXPOST method and data list in new page.	CEO318.1	CO318.1	BTL4
3	Create version control account on GitHub and using Git commands to create repository and push your code to GitHub.	CEO318.2	CO318.2	BTL4
4	Create Docker Container Environment (NVIDEIA Docker or any other).	CEO318.4	CEO318.4	BTL4
5	Create an Angular application which will do following actions: Register User, Login User, Show User Data on Profile Component	CEO318.3	CEO318.3	BTL4
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Lab Assignment 1A

Title:

Create a responsive web page which shows the ecommerce/college/exam admin dashboard with sidebar and statistics in cards using HTML, CSS and Bootstrap.

Objective:

To create a responsive web page that shows an e-commerce/college/exam admin dashboard. The dashboard should include a sidebar for navigation and statistics displayed in cards. This lab will make use of HTML, CSS, and Bootstrap framework.

Theory:

Write about HTML, CSS and Bootstrap.

Tools Required:

- Text editor (Visual Studio Code, Sublime Text, etc.)
- Browser (Google Chrome, Firefox, etc.)
- Bootstrap Framework(For Responsive Layout)

Pre-requisites:

 Basic understanding of HTML

 Basic understanding of CSS

 Familiarity with Bootstrap framework

Procedure:

1. **Create the Project Folder Structure:**

 Create a new folder for your project.

 Inside that folder, create an `index.html` file.

 Optionally, create a `style.css` file if you want to add custom styles.

2. **Link Bootstrap Framework:**

In your `index.html` file, include the Bootstrap CDN to leverage its responsive features. Add the following code inside the `<head>` tag to link Bootstrap's CSS and JS files:

```
<link
href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css" rel="
<script
src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
<script
src="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/js/bootstrap.min.js"></script>
```

3. HTML Structure:

Below is a basic structure for the dashboard:

```
<!DOCTYPE html>
<html>
<head>
<title>Page Title</title>

<link href="https://maxcdn.bootstrapcdn.com/bootstrap/4.5.2/css/bootstrap.min.css" rel="stylesheet">

</head>
<body>
<h1>My First Heading</h1>
<p>My first paragraph.</p>
</body>
</html>
```

Explanation:

- **Sidebar**: The sidebar is created using a div with the class sidebar. Inside it, we define anchor tags (<a>) for different sections of the dashboard (e.g., Home, Orders, Products, etc.).
- **Main Content Area**: The main content is displayed next to the sidebar, with some basic cards displaying statistics like Total Sales, Total Students, Upcoming Exams, and Pending Orders.
- **Cards**: Bootstrap's card component is used to display statistics in a visually appealing way. Cards are organized into columns using Bootstrap's grid system (col-md-3).

4. **Making the Sidebar Fixed and Responsive:** To make the sidebar fixed and ensure it stays on the left side while scrolling, you can use the position: fixed; property in CSS for the sidebar.

- Add the following CSS to the style section:

```
.sidebar {
position: fixed;
```

```
 width: 250px;
 top: 0;
 left: 0;
}
.main-content {
 margin-left: 270px;
}
```

5. **Testing the Responsive Layout:**

With Bootstrap, the layout will be responsive by default. However, you can test it by resizing the browser window

Conclusion:

Thus, from this assignment we have studied how to design responsive admin dashboard with sidebar for navigation, display key statistics in cards, and ensure the page is mobile-friendly using Bootstrap’s grid system.

Assignment 1B

Title:

Write a JavaScript Program to get the user registration data and push to array/local storage with AJAX POST method and data list in new page.

Objective:

The objective is to create a JavaScript program that collects user registration data through an HTML form and submits it using an AJAX POST method. The submitted data is stored in an array or local storage and displayed on a new page.

Implementation Steps

1. **Setup HTML Form**

Create a user registration form to collect details like name, email, and password. 2.

Validate Input Data

Ensure that user inputs are validated on the client side before submission (e.g., proper email format, required fields).

3. **AJAX POST Request**

Use JavaScript (or jQuery) to send the form data to a mock server endpoint via an AJAX POST request.

4. **Store Data**

Save the submitted data in an array or local storage for later use.

5. **Redirect and Display Data**

After successful registration, redirect the user to a new page where the stored data is retrieved and displayed.

 Purpose: This program demonstrates how to capture user input, send it to a server using AJAX, and manage data persistence with local storage.

 Technologies Used:

 HTML for creating the user interface.

 JavaScript for handling form submission, AJAX requests, and local storage operations.

 Local storage to persist data across sessions.

- **Workflow:**

 User fills out the registration form.

 JavaScript captures the data and validates it.

 Data is sent to a mock server endpoint using an AJAX POST request. On success, data is stored locally, and the user is redirected to a display page. The display page retrieves and presents the stored user data.

- **Learning Outcomes:**

 Understand client-side form validation.

 Learn how to use fetch for AJAX requests.

 Manage data using local storage.

 Dynamically update HTML content with JavaScript.

AJAX POST method:

The AJAX POST method is used to send data to a server asynchronously without reloading the web page. Unlike the GET method, it sends data in the request body, making it more secure and suitable for transmitting sensitive or large data. It is commonly used for tasks like submitting form data, creating or updating server resources, and interacting with APIs. The Content-Type header specifies the data format (e.g., JSON), and the server processes the request and responds with the result. The POST method enhances user experience by enabling smooth, dynamic updates.

Conclusion:

Thus, from this practical assignment, we able to learn how to implement user registration using the AJAX POST method in JavaScript. The program collects user input from an HTML form, validates the data, and sends it asynchronously to a server. The data is then stored in local storage and displayed on a new page.

Assignment 2A

Title:

Create version control account on GitHub and using Git commands to create repository and push your code to GitHub.

Objective

The objective of this practical assignment is to introduce students to version control by creating a GitHub account, initializing a local repository using Git, and pushing code to a remote repository on GitHub.

Steps to Complete the Practical

1. Create a GitHub Account

1. Visit GitHub and click on **Sign Up**.
2. Enter your details, including a valid email address, username, and password. 3. Complete the sign-up process by verifying your email address.

2. Install Git

1. Download Git from the official website.
2. Install it on your system by following the installation wizard for your operating system.

3. Set Up Git

Once Git is installed, configure it with your GitHub credentials using the following commands:

bash
git config --global user.name "Your GitHub Username"
git config --global user.email "your-email@example.com"

4. Initialize a Local Repository

1. Navigate to your project directory using the terminal or command prompt:
   bash
   cd /path/to/your/project

2. Initialize a Git repository:

bash

git init

3. Add all files in the directory to the staging area:

bash

git add .

4. Commit the files with a meaningful message:

bash

git commit -m "Initial commit"

5. Create a Repository on GitHub

Log in to your GitHub account.

Click the + icon in the top-right corner and select **New Repository**. Enter a repository name and choose whether it should be public or private. Click **Create Repository**.

6. Link Local Repository to GitHub

Copy the remote repository URL from GitHub.

In your terminal, link the local repository to the GitHub repository:

bash

git remote add origin <repository-URL>

Push your local code to GitHub:

bash

git push -u origin main

7. Verify the Push

Go to your GitHub repository page.

You should see the files from your local project uploaded to GitHub.

Common Git Commands

Command	**Description**
git init	Initializes a new Git repository.
git add .	Adds all files to the staging area.
git commit -m "message"	Commits changes with a message.
git remote add origin URL	Links the local repository to a remote one.
git push -u origin branch	Pushes changes to the specified branch.

Command	Description
git status	Shows the status of your repository.

Conclusion

From this practical assignment we able to learn how to use effectively Git and GitHub for version control. By completing these steps:

Initialize and manage a local Git repository.

Use Git commands to track changes and maintain versions.

Push their projects to a remote repository on GitHub for collaboration and backup. Mastering these skills is crucial for modern software development workflows and collaborative coding.

Assignment 2B

Title:

Create Docker Container Environment (NVIDEIA Docker or any other).

Objective:

The objective of this practical assignment is to set up a Docker container environment to run applications in an isolated, portable, and consistent environment. This can be achieved using any Docker distribution, including NVIDIA Docker for GPU-enabled applications. Steps to Create a Docker Container Environment

1. Install Docker

1.1 Download Docker:

Visit the Docker website and download the Docker Desktop version for your operating system.

1.2 Install Docker:

Follow the installation wizard for your OS (Windows, macOS, or Linux). On Linux, use the package manager to install Docker:

```
bash
sudo apt update
sudo apt install docker.io
```

1.3 Verify Docker Installation: Run the following command in your terminal or command prompt:

```
bash

docker --version
```

2. Install NVIDIA Docker (Optional for GPU Support)

If your application requires GPU acceleration (e.g., for AI/ML tasks), install NVIDIA Docker.

Install NVIDIA Driver: Ensure your system has the latest NVIDIA GPU driver installed.

Install NVIDIA Container Toolkit:

```
bash
distribution=$(. /etc/os-release;echo $ID$VERSION_ID)
curl -s -L https://nvidia.github.io/nvidia-docker/gpgkey | sudo gpg --dearmor -o
/usr/share/keyrings/nvidia-container-toolkit-keyring.gpg
curl -s -L https://nvidia.github.io/nvidia-docker/$distribution/nvidia-docker.list | sudo  tee
/etc/apt/sources.list.d/nvidia-docker.list
```

sudo apt update

sudo apt install -y nvidia-container-toolkit

sudo systemctl restart docker

Verify NVIDIA Docker Installation:

bash

docker run --rm --gpus all nvidia/cuda:12.0-base nvidia-smi

3. Create a Docker Container Environment

Pull a Docker Image: Choose an image to use as the base environment. For example:

bash

docker pull ubuntu:latest

This command pulls the latest Ubuntu image.

Run a Docker Container: Create and start a container from the pulled image:

bash

docker run -it --name my-container ubuntu:latest

-it: Interactive terminal.

--name: Assigns a name to the container.

Install Dependencies in the Container: Once inside the container, install any required software. For example:

bash

apt update

apt install -y python3

Save the Customized Container: After configuring the container, save it as a new image:

bash

docker commit my-container my-custom-image

4. Run the Custom Docker Container

Start the container from the saved image:

bash

docker run -it my-custom-image

List all running containers:

bash

docker ps

Stop a running container:

bash

docker stop <container_id>

Remove unused containers and images:

bash

docker rm <container_id>

docker rmi <image_name>

5. Use Docker Compose for Multi-Container Applications

For complex environments with multiple services (e.g., web server, database), use Docker Compose.

Install Docker Compose:

Docker Desktop comes with Docker Compose pre-installed.

On Linux, install it with:

bash

sudo apt install docker-compose

Create a docker-compose.yml **File**: Define your services and configurations in this file:

yaml

version: "3.9"

services:

web:

image: nginx:latest

ports:

- "8080:80"

database:

image: mysql:latest

environment:

MYSQL_ROOT_PASSWORD: password

Run Docker Compose: Start all services with:

bash

docker-compose up

Conclusion

This practical assignment demonstrates how to create a Docker container environment for isolated and consistent application deployment.

Assignment 2C

Title:

Create an Angular application which will do following actions: Register User, Login User, Show User Data on Profile Component

Objective

The objective of this assignment is to create an Angular application that allows users to register, log in, and view their profile information. The application will include three core functionalities: **User Registration**, **User Login**, and **Profile Display**.

Write explanation about AngularJS

Conclusion:

From this practical assignment we able to demonstrates the creation of an Angular application with essential user management features